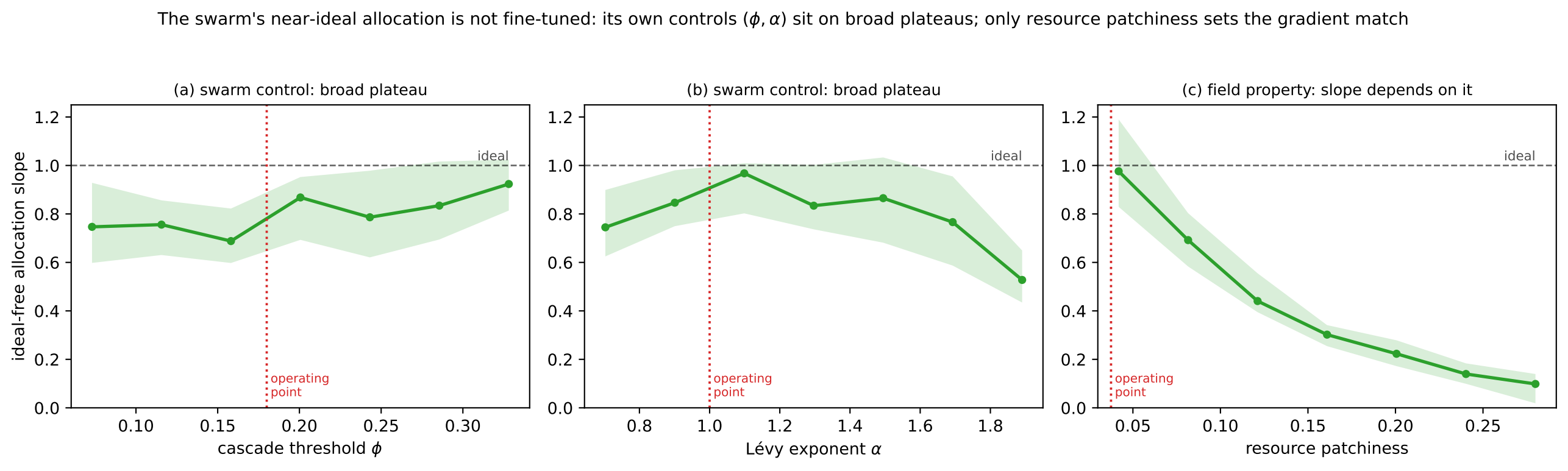
The swarm's near-ideal allocation is not fine-tuned: its own controls (ϕ, α) sit on broad plateaus; only resource patchiness sets the gradient match
(a) swarm control: broad plateau
(b) swarm control: broad plateau
(c) field property: slope depends on it
ideal-free allocation slope
ideal
operating point
cascade threshold ϕ
Lévy exponent α
resource patchiness
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.10
0.15
0.20
0.25
0.30
0.8
1.0
1.2
1.4
1.6
1.8
0.05
0.10
0.15
0.20
0.25